Operational Health Metrics

Date: January 10, 2025 **Report Period:** Last 90 Days

Summary

> This document provides operational health and uptime metrics for all three components of the Crypto-AI-Bot SaaS platform: the core bot, the Signals-API gateway, and the Signals-Site frontend.

Infrastructure Overview

Deployment Architecture

Compute: Fly.io (3 apps)

- `crypto-ai-bot` — Core intelligence engine
- `crypto-signals-api` — API gateway and middleware
- `signals-site` — Next.js frontend (not yet deployed to Fly)

Data Store: Redis Cloud (managed instance)

- Host: `redis-19818.c9.us-east-1-4.ec2.redns.redis-cloud.com:19818`
- Connection: TLS-encrypted with CA cert
- Plan: Shared instance (1GB memory, ~500 connections)

CDN/Hosting (Frontend): Vercel (planned for signals-site)

Infrastructure Diagram

Uptime & Availability

Crypto-AI-Bot (Core Engine)

Last 90 Days:

- **Uptime:** 99.2%
- **Downtime Events:** 3 planned restarts, 2 Redis connection drops
- **Mean Time to Recovery:** <5 minutes

Key Metrics:

- Average CPU: 45-60%
- Memory Usage: 512MB (1GB allocated)
- Restart Policy: Automatic on failure, exponential backoff

Signals-API (Gateway)

Last 90 Days:

- **Uptime:** 99.7%
- **API Latency (p50):** 18ms
- **API Latency (p95):** 85ms
- **API Latency (p99):** 220ms

Health Check Endpoint: `GET /v1/signals/health`

- Response: `{"status": "healthy", "redis": "connected", "uptime_sec": 123456}`

API Latency Distribution

Signals-Site (Frontend)

Status: In development, not yet deployed to production.

- Local testing: Stable
- Planned deployment: Vercel (auto-scaling, global CDN)

Redis Cloud Health

Connection Stability

Last 90 Days:

- **Connection Uptime:** 99.8%
- **Dropped Connections:** 4 (all recovered within <30s)
- **TLS Errors:** 0

Connection Pool:

- Max connections: 50
- Active connections (peak): 12-18
- Idle timeout: 300s

Memory & Performance

- **Memory Used:** 420MB / 1GB (42%)
- **Keys:** ~5,000 active keys
- **Eviction Policy:** `allkeys-lru`
- **Hit Rate:** 94%

Redis Memory Usage

Monitoring & Alerting

Prometheus Metrics

Exported Metrics:

- `bot_signal_latency_ms` — Signal generation latency
- `bot_trade_count_total` — Total trades executed
- `bot_pnl_usd` — Cumulative PnL (paper trading)
- `bot_redis_connection_status` — Redis health (0/1)

- `api_request_duration_seconds` — API response times
- `api_request_count_total` — Total API requests

Scrape Interval: 15s

Grafana Dashboards

1. **Paper Trial Dashboard** — Real-time trading performance
2. **Maker Monitoring Dashboard** — Order book and execution quality
3. **PnL Aggregation Dashboard** — Cumulative returns by pair and strategy

Prometheus Metrics Overview

Error Rates & Incident Log

Error Rates (Last 30 Days)

Crypto-AI-Bot:

- Fatal errors: 0
- Redis connection errors: 2 (both auto-recovered)
- Kraken WebSocket disconnects: 8 (all reconnected <10s)

Signals-API:

- 5xx errors: 0.02% (14 out of ~70,000 requests)
- 4xx errors: 0.15% (mostly invalid query params)
- Timeout errors: 0

Root Causes:

- 5xx errors: Redis connection lag during high traffic
- Kraken disconnects: Upstream provider maintenance

Incident Log

Date	Component	Issue	Duration	Resolution
2024-12-15	Bot	Redis connection timeout	45s	Auto-reconnect
2024-12-28	API	Fly.io region outage	8m	Failover to secondary
2025-01-03	Bot	Kraken WebSocket rate limit	2m	Backoff + retry

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Deployment & Release Cadence

Deployment Frequency

Last 90 Days:

- Bot: 18 deployments (avg 1-2 per week)
- API: 12 deployments
- Site: N/A (not yet in production)

Deployment Method:

- Fly.io: `fly deploy` (Docker-based, rolling updates)
- GitHub Actions: CI/CD pipeline with automated tests
- Rollback capability: `fly releases revert`

Release Strategy

1. Feature development on `feature/*` branches
2. Merge to `main` after PR review
3. Automated tests run via GitHub Actions
4. Manual deployment to Fly.io (staged rollout)
5. Monitoring for 24h post-deployment

Performance Benchmarks

Signal Generation Performance

- **Signals/Minute (Peak):** ~50
- **Signals/Day (Average):** ~2,000-3,000
- **CPU per Signal:** <10ms
- **Memory per Signal:** ~2KB

API Throughput

- **Requests/Second (Peak):** 45
- **Requests/Day (Average):** ~15,000
- **Concurrent Connections:** Up to 20 SSE streams

API Throughput Graph

Scalability Assessment

Current Capacity

- **Bot:** Can handle 10+ pairs simultaneously (currently 5)
- **API:** Can serve 100+ concurrent SSE clients
- **Redis:** 1GB sufficient for 6-12 months at current growth

Scaling Plan

Vertical Scaling (Short-term):

- Increase Fly.io machine size (1GB → 2GB RAM)
- Upgrade Redis Cloud plan (1GB → 5GB)

Horizontal Scaling (Long-term):

- Deploy multiple bot instances (per-pair sharding)

- Load-balance API across 2+ regions
- Implement Redis Cluster for higher throughput

Compliance & Security

Security Measures

- **TLS Encryption:** All Redis connections, all API endpoints
- **Secrets Management:** Fly.io secrets, no hardcoded credentials
- **Access Control:** Redis password auth, API key validation (planned)

Backup & Disaster Recovery

- **Redis Snapshots:** Daily automatic backups (Redis Cloud)
- **Code Repositories:** GitHub (private repos, 2FA enabled)
- **Deployment Rollback:** One-command rollback via Fly.io

End of Document